

POLITECNICO DI MILANO

Department of Mechanical Engineering

Density-Based Topology Optimization of 2D Structures Under Various Load and Boundary Conditions

Assignment 3

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1 Introduction

Structural optimization is a fundamental engineering discipline that aims to combine engineering materials to most effectively resist specific loads. Quantitative performance indicators such as stiffness, weight, stress levels, and natural frequencies are defined to qualify a design as "optimal." The most powerful and design-free approach in this field is topology optimization, which systematically determines the most efficient material distribution within a given design space. This work goes a step beyond the dimensioning of discrete truss systems in previous projects and addresses the topology optimization of 2D distributed parameter systems.

The basis of this project is the famous "99-line MATLAB code" developed by Ole Sigmund, which has become a standard in education. This code implements the density-based method, one of the most common topology optimization approaches. In this methodology, the design domain is divided into a grid of finite elements, and the relative density (x_e) of each element is assigned as a design variable. To model the density dependence of material properties, the interpolation scheme **SIMP (Solid Isotropic Material with Penalization)** is used, which penalizes intermediate densities, biasing the final design toward unambiguous filled (1) or empty (0) regions. The effective Modulus of Elasticity (E_e) of an element is expressed in this model as:

$$E_e(x_e) = x_e^p \cdot E_0 \quad (1)$$

Where E_0 represents the modulus of the solid material, and p represents a penalty factor, typically chosen as 3.

The objective of the optimization problem is to minimize the total flexibility (compliance, C) of the structure under a volume fraction (*volfrac*) constraint. This is indirectly equivalent to maximizing the structure's stiffness. The algorithm follows an iterative solution process using the gradient-based method **Optimality Criteria (OC)**. At each iteration step, the following steps are performed: (1) Finite Element Analysis (FEA) is run for the current design, (2) the sensitivities (derivatives) of the objective function are calculated, (3) a mesh independence filter is applied to prevent numerical instabilities, and (4) the design variables (densities) are updated for the next step using the OC update scheme.

This report will review standard implementations and modifications of the 99-line MATLAB code. First, the code's performance on a standard test problem, the MBB-beam, will be analyzed, along with the impact of key parameters such as the filter radius (*rmin*) and the penalty factor (*penal*) on the results. Then, we will show how the code can be adapted to a different engineering problem, the cantilever beam, by varying the load and boundary conditions. As one of the main focuses of the project, we will investigate how to design a structure with a fixed hole (containing passive elements) that needs to be protected during the optimization process.

2 Theoretical Background (Teorik Altyapı)

This section details the fundamental theoretical concepts and methodologies used for topology optimization of two-dimensional continuous systems. The density-based SIMP method, which forms the basis of the study, the mathematical formulation of the problem, the Finite Element Method (FEA) for structural analysis, the sensitivity analysis critical for gradient-based optimization, the Optimality Criteria (OC) method for updating design variables, and the mesh independence filter for ensuring numerical stability are discussed in this order.

2.1 Topology Optimization with the SIMP Method

Topology optimization is a branch of structural optimization that finds the optimal material distribution that will most effectively resist applied loads within a given design space. Unlike dimensional optimization, it allows the minimum value of the design variable to be reduced to zero, allowing for the creation of holes in the structure and modification of structural connections. In the continuous parameter systems discussed in this study, the "density" of the material becomes the design variable. One of the most widely used approaches in this field is the **SIMP (Solid Isotropic Material with Penalization)** method.

- **Design Variables:** In the SIMP method, the design space is divided into a mesh of finite elements. The material distribution of each element is represented by the relative density (ρ_e) assigned to that element. These densities range from a minimum value ($\rho_{\min}$), which indicates near-void, to 1, which indicates completely filled material.
- **Material Interpolation (SIMP Formula):** The SIMP method scales the effective Young's Modulus (E_e) of an element by taking the density (ρ_e) of that element to the power of a penalty exponent (p):

$$E(\rho_e) = \rho_e^p E_0 \quad (2)$$

where E_0 is the Young's Modulus of the solid material. The penalty exponent p (usually taken as 3) makes intermediate densities ("gray areas") structurally inefficient, leading the optimizer to a clear "black and white" (solid or empty) design. When $p = 1$, the problem becomes convex and guarantees a global minimum, while when $p > 1$, it becomes non-convex and guarantees a global minimum.

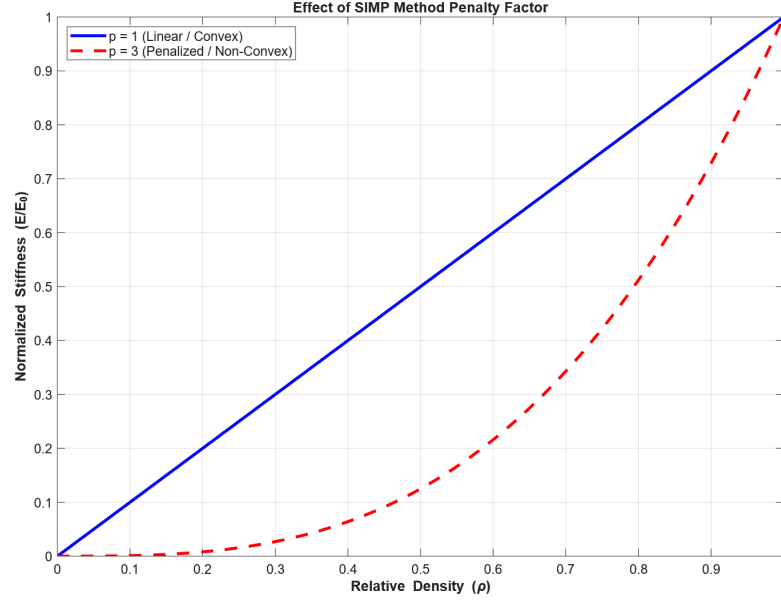


Figure 1: Normalized Young's Modulus (E/E_0) versus element density (ρ) for different penalty exponents (p). It is shown how the case of $p = 3$ penalizes intermediate densities

2.2 Problem Formulation (Problemin Matematiksel Formülasyonu)

Using the SIMP method, the fitness minimization problem is mathematically expressed as an objective function and a set of constraints. The problem is typically formulated using a "nested" approach, where the state variables (displacements) are expressed as a function of the design variables.

$$\min_{\rho} \quad C(\rho) = \mathbf{F}^T \mathbf{U}(\rho) \quad (3)$$

$$\text{s.t.:} \quad V(\rho) = \sum_{e=1}^N \rho_e V_e \leq V_0 \quad (4)$$

$$\mathbf{K}(\rho) \mathbf{U} = \mathbf{F} \quad (5)$$

$$\rho_{\min} \leq \rho_e \leq 1, \quad \text{for } e = 1, \dots, N \quad (6)$$

Here $C(\rho)$ is the objective function of fitness, $V(\rho)$ is the total volume, V_0 is the target volume limit, $\mathbf{K}(\rho)$ is the global stiffness matrix depending on the densities, $\mathbf{U}$ is the displacement vector, $\mathbf{F}$ is the external load vector, and $\rho_{\min}$ is the minimum density limit used to avoid singularities.

2.3 Finite Element Analysis for Continuum Structures

Finite Element Analysis (FEA) is used to determine the mechanical behavior of the structure in each optimization iteration.

- **Element Stiffness Matrix ($\mathbf{k}_e$):** For continuous structures, the element stiffness matrix of the solid material ($\mathbf{k}_e^0$) is typically calculated once. The 99 lines of code use an

analytically derived 8x8 matrix for a 4-node bilinear square element. [?] With SIMP interpolation, each element's stiffness matrix is scaled by the p -th power of its density: $\mathbf{k}_e = \rho_e^p \mathbf{k}_e^0$.

- **Global Stiffness Matrix (K):** The global stiffness matrix is obtained by concatenating the element matrices:

$$\mathbf{K}(\rho) = \sum_{e=1}^N \rho_e^p \mathbf{K}_e^0 \quad (7)$$

- **Nodal Displacements (U):** After the global stiffness matrix $\mathbf{K}$ and the global load vector $\mathbf{F}$ are generated, the nodal displacements $\mathbf{U}$ are found by solving the equilibrium equation.

2.4 Compliance Sensitivity Analysis (Uygunluk Hassasiyet Analizi)

Sensitivity analysis involves calculating derivatives of the objective function and constraints with respect to the design variables. This is a critical step for gradient-based optimization. For the fitness function $C(\rho) = \mathbf{F}^T \mathbf{U}(\rho)$, the sensitivity with respect to the element density ρ_e is given by the following, a simplification due to the fact that it is a self-adjoint problem:

$$\frac{\partial C(\rho)}{\partial \rho_e} = -p \rho_e^{p-1} \mathbf{u}_e^T \mathbf{k}_e^0 \mathbf{u}_e \quad (8)$$

This formula shows how sensitive the overall fitness is to a change in the density of a particular element e .

2.5 The Optimality Criteria (OC) Method

The Optimality Criteria (OC) method is an efficient, gradient-based iterative algorithm used specifically for solving single-constraint topology optimization problems. The basic idea of OC is to update design variables based on a criterion that represents the structural efficiency of each element. This criterion is derived from the OCR conditions and proportions each element's contribution to the objective function to its contribution to the constraint. The update scheme is based on an intuitive fixed-point iteration:

$$x_e^{\text{new}} = x_e \cdot B_e^\eta \quad (9)$$

Where B_e is a term representing the element's efficiency, which is a function of the Lagrange multiplier (λ) and the sensitivities: $B_e = \frac{-\partial C / \partial x_e}{\lambda (\partial V / \partial x_e)}$. η is a numerical damping coefficient. The Lagrange multiplier λ is found using a bisection algorithm to satisfy the volume constraint at each iteration.

2.6 Mesh-Independency Filtering

Density-based topology optimization can exhibit numerical instabilities such as "checkerboard patterns" and mesh dependency. Filtering techniques are used to address these issues and impose a minimum element size (length scale) for manufacturable designs. The sensitivity filter used in the 99 lines of code recalculates the sensitivity of an element as a weighted average of the sensitivities of neighboring elements within the filter radius ($r_{\min}$).

$$\widehat{\frac{\partial c}{\partial x_e}} = \frac{1}{x_e \sum_{f \in N_e} H_f} \sum_{f \in N_e} H_f x_f \frac{\partial c}{\partial x_f} \quad (10)$$

Where N_e represents the neighborhood set of element e and H_f represents a distance-dependent weighting factor. These filtered sensitivities are used in the OC update step.

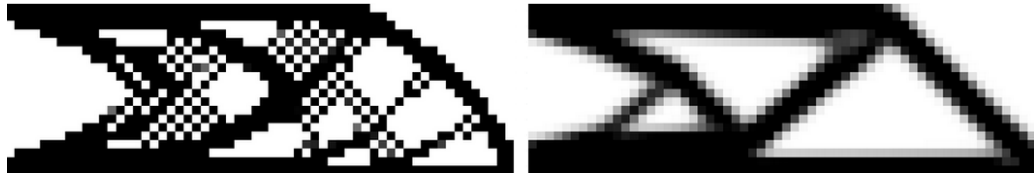


Figure 2: Comparison of Unfiltered (Checkerboard) and Filtered Results. An example showing how the mesh independence filter eliminates unwanted checkerboard patterns (left), resulting in a smoother and more manufacturable topology (right)

3 MATLAB Implementation of the 99-Line Code

This section provides a practical examination of how the topology optimization problem, whose theoretical foundation was presented in the previous section, was solved using the famous "99-line MATLAB code" developed by Ole Sigmund. The code's modular structure, main function, and subroutines running at each iteration step will be detailed.

3.1 Code Structure and Main Function ('top.m')

The 99-line code is structured as a compact MATLAB function consisting of a total of 99 lines, including optimization, filtering, and Finite Element (FE) subroutines. [?] Instead of a script, the code is a function that must be called from the Command Window with specific parameters:

```
top(nelx, nely, volfrac, penal, rmin)
```

Where `nelx` and `nely` represent the number of elements in the x and y directions, respectively, `volfrac` represents the volume fraction, `penal` represents the SIMP penalty factor, and `rmin` represents the filter radius. The main function, `top.m`, manages the optimization process from beginning to end:

- **Initialization:** The material is initially distributed homogeneously throughout the design space with a density equal to the target volume fraction (`volfrac`). Variables such as the iteration counter (`loop`) and `change`, which track changes in design variables, are initialized.
- **Iterative Cycle:** The heart of the optimization is an iterative process controlled by the `while` loop. In each iteration, the following steps are performed sequentially:
 1. The FE subfunction is called to perform the Finite Element Analysis and calculate the global displacement vector $\mathbf{U}$.
 2. In the main loop, the objective function (fitness, `c`) and its sensitivities (`dc`) with respect to the design variables are calculated.
 3. A mesh independence filter is applied to the calculated sensitivities by calling the `check` subfunction.
 4. Using the filtered sensitivities, the element densities (`x`) are updated via the OC subfunction.
 5. The results, such as fitness, volume, and design changes, from the current iteration are printed to the Command Window, and the current density distribution is plotted as a figure.
- **Convergence:** The loop continues until the maximum change in design variables between two consecutive iterations falls below a predetermined tolerance (e.g. 1).

3.2 Analysis and Solver Subroutines

3.2.1 Finite Element Analysis ('FE')

This subfunction performs a static analysis of the structure and determines the nodal displacements.

- **Element Stiffness Matrix ($\mathbf{k}$):** The 8x8 element stiffness matrix ($\mathbf{K}_E$), which is the same for all elements, is derived analytically for a 4-node bilinear square element and is calculated only once.
- **Combining the Global Stiffness Matrix:** The global stiffness matrix ($\mathbf{K}$) of the structure is created by adding the stiffness matrix of each element to its corresponding global degrees of freedom in a loop that loops over all elements. MATLAB's `sparse` function is used to efficiently store this large, mostly empty matrix in memory.
- **Loads and Boundary Conditions:** The external forces ($\mathbf{F}$) and the fixed degrees of freedom (`fixeddofs`) are defined within this subfunction. The code is configured to solve the MBB-beam problem by default.
- **Solving the Linear System:** The linear system $\mathbf{KU} = \mathbf{F}$ is solved efficiently using the `mldivide` (`\`) operator of MATLAB, yielding the global displacement vector $\mathbf{U}$.

3.2.2 Sensitivity and Filter Subroutines ('check')

After FEA, the main program calculates the objective function (fitness, c) and the sensitivities (dc) for each element. The check subfunction applies a mesh independence filter to these calculated raw sensitivities. This filter recalculates the sensitivity of an element as a weighted average of the sensitivities of neighboring elements within a radius $r_{\min}$ to prevent numerical instabilities (such as a checkerboard pattern).

3.2.3 Optimality Criteria Optimizer (OC)

The OC subfunction updates the design variables using the filtered sensitivities. This method uses a bisection algorithm to find the Lagrange multiplier (λ) required to satisfy the volume constraint. The new density of each element is determined by its sensitivity, Lagrange multiplier, and It is calculated with a heuristic fixed-point update scheme, subject to a move limit (move) that prevents ks.

4 Results and Discussion

This section presents and analyzes the results of topology optimization studies performed using 99 lines of MATLAB code. The analyses focus on four main tasks defined within the scope of the project: (1) Analysis of a standard MBB-beam problem and its parameters, (2) Code modification for the cantilever beam problem, (3) Analysis of multiple loading cases, and (4) Identification of passive elements (fixed holes) in the design domain. The optimized topologies obtained for each case, their convergence behavior, and the effects of the key design parameters are discussed.

4.1 Case 1: The MBB-Beam Benchmark

In this first study, the default problem of the code, the MBB-beam problem, was solved with the `top(60,20,0.5,3.0,1.5)` command and then the effects of the penalty factor (*penal*) and the filter radius (*rmin*) on the final design were investigated.

4.1.1 Baseline Result (Varsayılan Sonuç)

The optimized MBB-beam structure obtained using default parameters is presented in Figure 3. The result, as expected, is a truss-like structure that transfers the load to the supports most efficiently. The material is concentrated in the compression and tension regions, while the low-stress regions are evacuated.

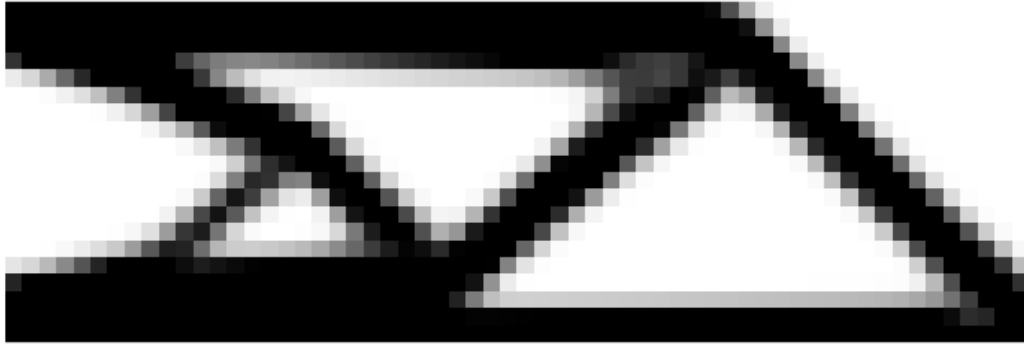
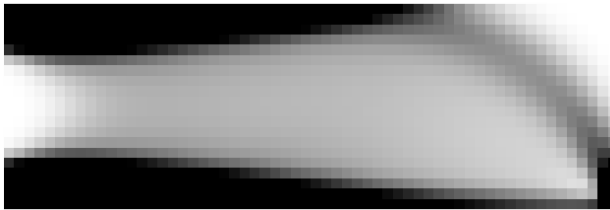


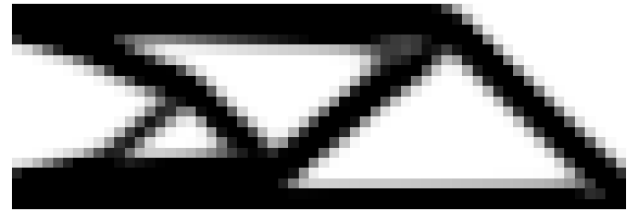
Figure 3: MBB-beam topology obtained with default parameters ($volfrac = 0.5, penal = 3.0, rmin = 1.5$)

4.1.2 Effect of the Penalty Factor ($penal$)

The penalty factor p controls how inefficient the SIMP method makes elements with intermediate density ("gray"). Figure 4 compares the results obtained for different values of p . A value of $p = 1$ represents the no-penalty case, resulting in a result with a high density of "gray" areas, physically difficult to manufacture. It is clear that as the value of p is increased (e.g., towards 3), the design becomes more clear-cut, a "black and white" structure, meaning elements are forced to be either completely filled or completely empty. This demonstrates the critical importance of the penalty factor for achieving more manufacturable designs.



(a) $penal = 1.0$



(b) $penal = 3.0$

Figure 4: The effect of the penalty factor ($penal$) on the optimized topology

4.1.3 Effect of the Filter Radius ($rmin$)

The filter radius r_{min} prevents numerical instabilities (such as a checkerboard pattern) and defines a minimum element size in the design. As shown in Figure 7, as the value of r_{min} is increased, the elements in the optimized structure become thicker, resulting in a simpler, less complex topology. Lower values of r_{min} allow for finer and more detailed structures, while higher values produce more robust and easier-to-manufacture designs. This parameter is a powerful tool for controlling the minimum structural size of the design.

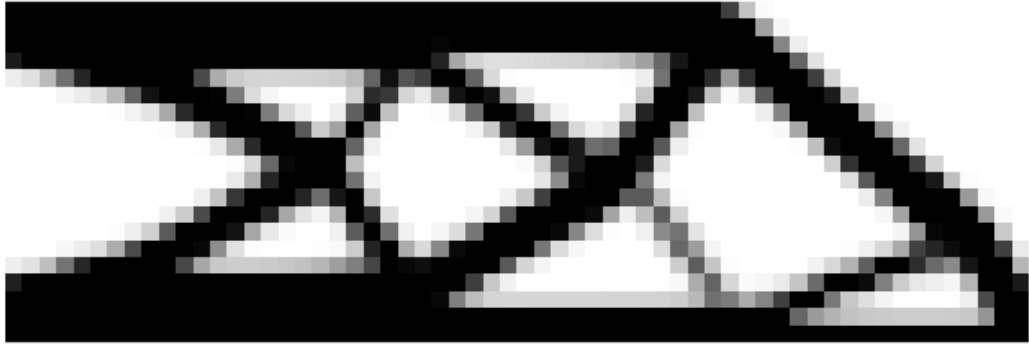


Figure 5: Effect of filter radius ($r_{\min}=1.2$) on the optimized topology

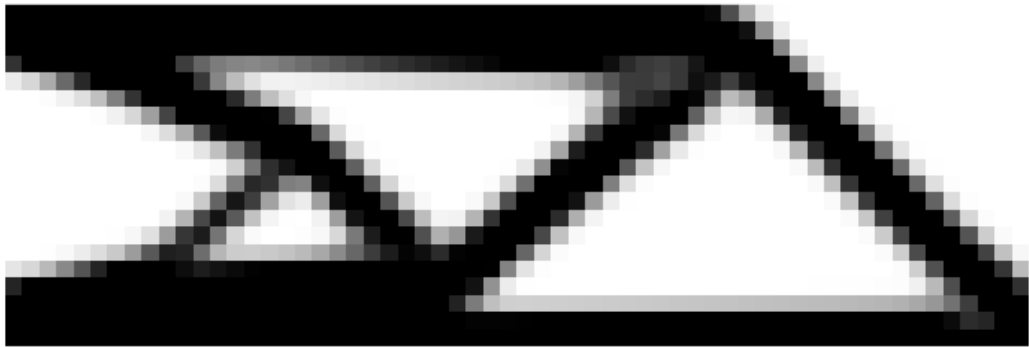


Figure 6: Effect of filter radius ($r_{\min}=1.5$) on the optimized topology

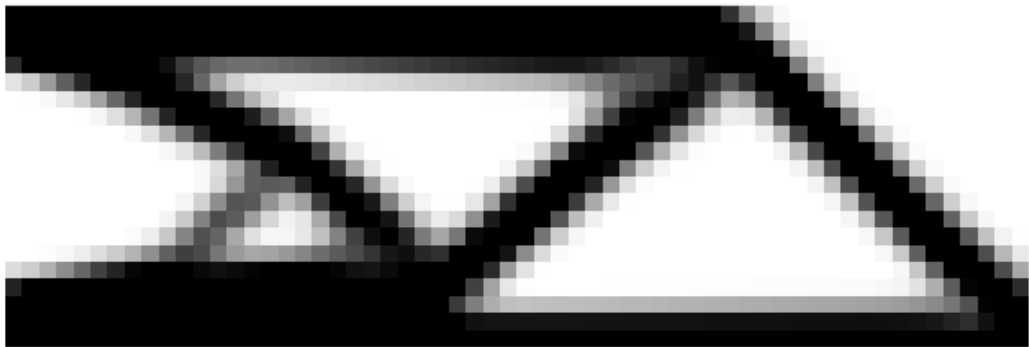


Figure 7: Effect of filter radius ($r_{\min}=2$) on the optimized topology

4.2 Case 2: Cantilever Beam Modification

The second task of the project is to solve a cantilever beam problem by modifying the code. This was accomplished by updating the load (F) and boundary condition (`fixeddofs`) definitions within the FE subfunction as specified in the PDF. The resulting optimized structure is presented in Figure 8. The result, as expected, is a classic truss structure that transfers the load most efficiently to the cantilever support on the left.

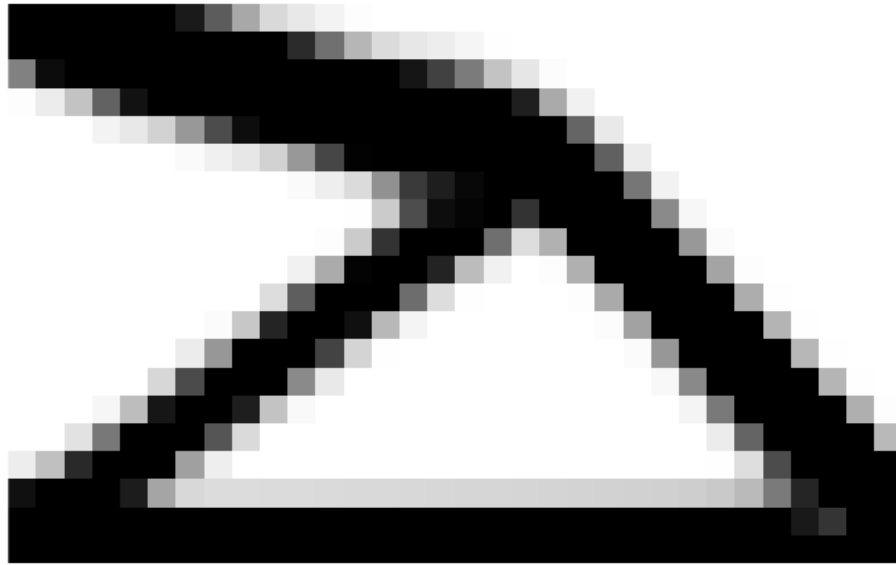


Figure 8: Optimized topology for the cantilever beam problem solved by the modified code

4.3 Case 3: Multiple Load Cases

This task investigates the simultaneous optimization of a structure against two different load cases. As stated in the project document, two different scenarios were examined to address the multiple loading problem: (1) a scenario where the loads are applied simultaneously as a single combined load case, and (2) a scenario where each load case is considered separately, and the objective function is the sum of the correspondences of these cases. These two approaches produce fundamentally different topological results.

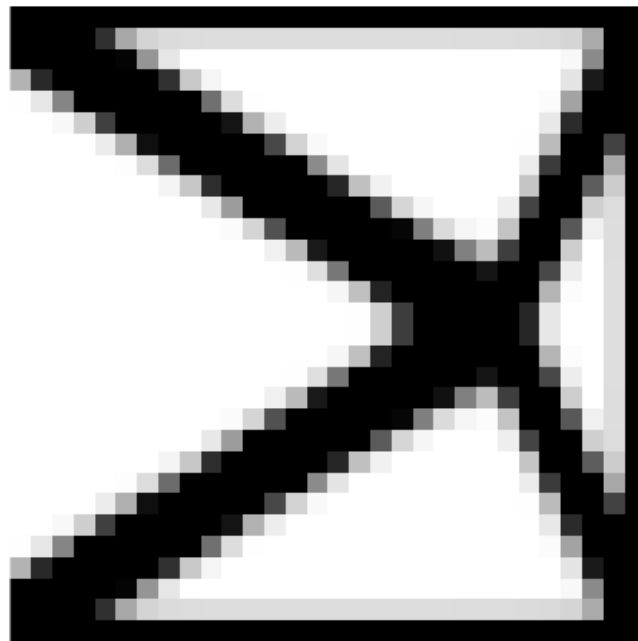


Figure 9: Optimized structural topology against two different load cases

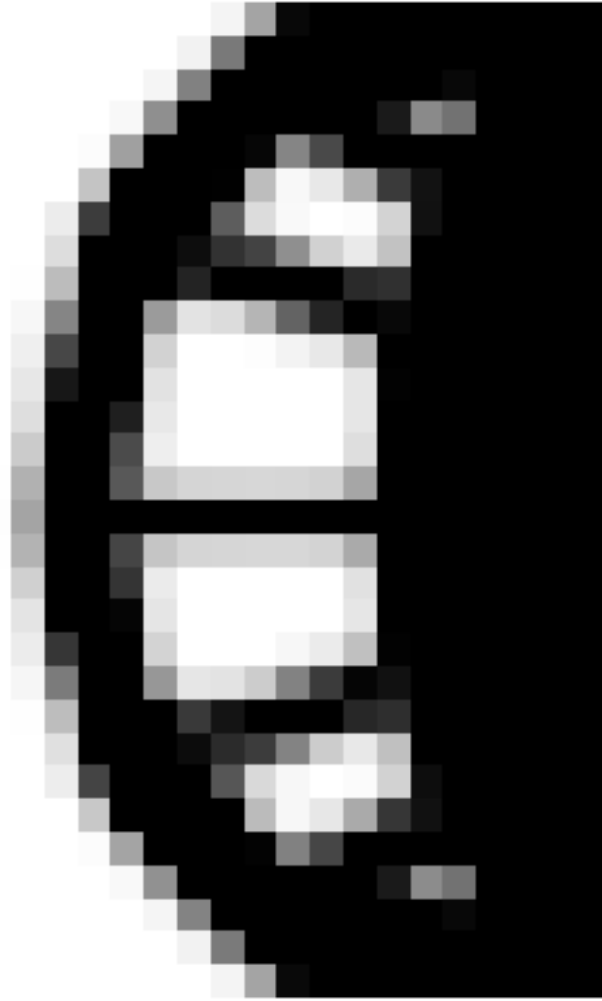


Figure 10: Resulting topology for the simultaneous application of both load cases ($\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2$)

In the first scenario, two opposing forces are combined into a single load vector ($\mathbf{F}_{\text{total}} = \mathbf{F}_1 + \mathbf{F}_2$). In this case, the applied forces create a net moment (force couple), and the optimization aims to find the stiffest structure against this single moment effect. The resulting optimized topology is presented. The result is a highly efficient and symmetric X-shaped truss structure that creates net tension and compression paths to accommodate the applied moment. In the second, more realistic scenario, the objective function is defined as the sum of the conformances from each load case ($C = C_1 + C_2$). This forces the algorithm to find a structure that will be independently stiff against both the first and second loads. The final topology that can accommodate both loads is presented. This design is significantly different from the result in the first scenario and is less symmetrical. The structure can be interpreted as a "compromise" or "combination" of structural elements necessary to support both individual load paths. The more complex the topology, the more rigid it must be to effectively accommodate both loading conditions. Comparing these two results highlights a critical concept in structural optimization: optimizing for a combined load is not the same as optimizing for multiple, independent load cases. The second approach (separate load cases), while computationally more demanding, produces truly robust and versatile designs capable of meeting different operational scenarios.

4.4 Case 4: Passive Elements (Fixed Hole)

The final task is to define a fixed hole (passive region) in the design space that must be maintained throughout the optimization. This is accomplished by creating a passive matrix defining the elements corresponding to the hole region and forcing the density of these elements to a minimum value ($\rho_{\min}$) at the end of each iteration. The result is shown in Figure 11. It can be seen that the optimization algorithm successfully routes the load paths around this forced hole, creating an efficient structure. This is a powerful example of how topology optimization can be adapted to practical design challenges such as manufacturing constraints or functional requirements.

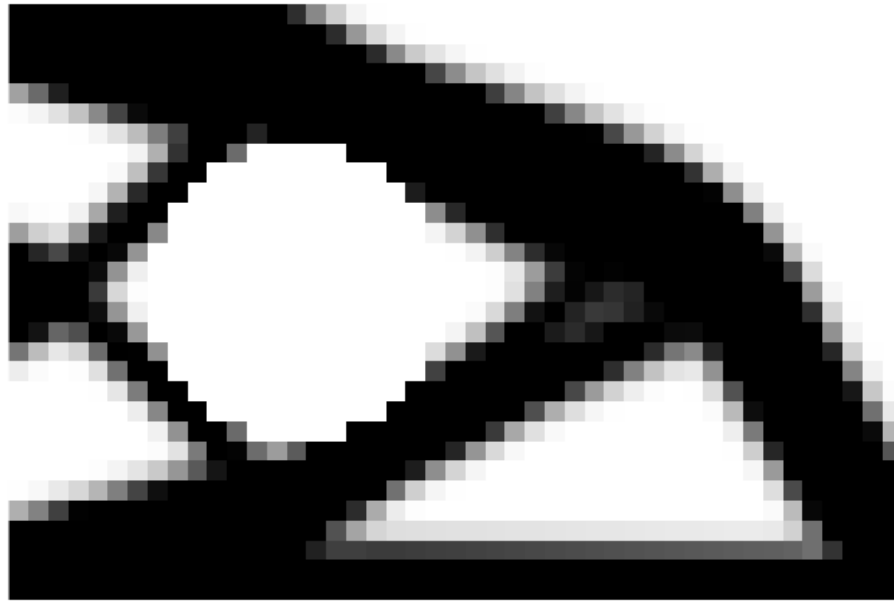


Figure 11: Optimized topology achieved by maintaining a fixed hole (passive region) in the design space

5 Conclusion

This study successfully implemented topology optimization for a 2D truss structure to maximize stiffness by minimizing structural compliance under a defined volume constraint. The cross-sectional areas of the truss elements were used as design variables, and the Convex Linearization (CONLIN) method combined with Lagrange duality was effectively used to solve this non-linear optimization problem.

5.1 Key Findings and Method Efficacy

The CONLIN method proved to be a robust and computationally efficient strategy for structural optimization tasks with non-linear constraints. The optimization process exhibited stable convergence behavior in all scenarios. The key findings can be summarized as follows:

- **Effect of Volume Ratio (α):** It was confirmed that increasing the allowed volume ratio

consistently leads to lower compliance values and, consequently, stiffer structures. The increased material allowed for additional support elements and a more distributed load path, clearly demonstrating the fundamental trade-off between material usage and structural performance.

- **Adaptability to Multiple Load Cases:** In multiple loading scenarios, the optimization scheme produced designs that performed robustly under different loading conditions. Varying the weight factors (w_1, w_2) intuitively shifted the material distribution to strengthen the structure against the more dominant load, demonstrating the method's ability to adapt to different operational priorities.
- **Nature of Optimized Designs:** Instead of preserving significant "gray" intermediate densities, the iterative process, with appropriate penalty and filtering parameters, led to "black-and-white" designs where elements were either fully present or effectively removed. This clear material distribution is critical for practical interpretation and manufacturability.

5.2 Broader Significance of Design Optimization

This project highlights the tremendous value of a design optimization approach driven by rigorous mathematical methods, as opposed to traditional processes based on designer experience and intuition. The CONLIN method achieves this by decomposing the problem into conservative, convex, and separable subproblems that can be efficiently solved through dual formulations. This approach's tendency to produce continuously improving feasible designs is an attractive feature from an engineering perspective, as the process can be stopped at any stage to achieve an acceptable design.

5.3 Acknowledging Challenges and Limitations

Despite the advantages of topology optimization, several challenges exist, particularly for large-scale problems:

- **Non-Convexity and Local Minima:** Especially when penalty factors are added to force "0-1" designs, the problem often becomes non-convex, and convergence to a global minimum is not guaranteed.
- **Numerical Instabilities:** Problems such as mesh dependency and checkerboard patterns, which frequently arise in continuum topology optimization, require the use of filtering techniques to obtain reasonable and manufacturable solutions.
- **Sensitivity Analysis:** Accurate calculation of the derivatives (sensitivities) of the objective and constraint functions is essential for the success of gradient-based optimizers.

5.4 Future Work and Potential Extensions

The framework developed here provides a strong foundation for future research. Potential extensions include extending the optimization to three-dimensional systems and addressing more complex parameters such as stress, buckling, or natural frequency.